

Himan Abdollahpouri

Senior Research Scientist | Personalization | Recommender Systems

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SUMMARY:

Senior Research Scientist specializing in machine learning, recommender systems, and long-term user value optimization. Led research in multi-objective ranking, bias mitigation, causal inference, and sequential recommendation, with recent work on foundation models and LLM-based evaluation using human feedback. Partnered with engineering to translate research into large-scale production personalization systems. Author of 30+ publications (4k+ citations, h-index 24).

TECHNICAL SKILLS

- **Recommender Systems:** Multi-objective & long-term optimization, contextual bandits, reward modeling
- **LLM Evaluation:** LLM-as-judge systems, human feedback fine-tuning
- **Causal & Experimental Methods:** Causal inference, heterogeneous treatment effects (HTE), A/B testing, off-policy evaluation
- **Tools:** Python, PyTorch, TensorFlow, SQL

EDUCATION & TRAINING

University of Colorado Boulder — Ph.D in Computer Science (2015–2020)

Dissertation: Popularity bias in recommender systems and its multi-stakeholder impact

Northwestern University — Postdoctoral Fellow (2020–2021)

Research: Personalization and multi-objective recommendation in news and media

WORK EXPERIENCE

Spotify, Chicago - *Senior Research Scientist*

September 2021 - PRESENT

- Led development of an LLM-as-judge framework using human feedback, adopted by multiple product teams for offline evaluation.
- Led research on reward modeling for long-term optimization, with resulting proxy metrics adopted across teams as optimization targets.
- Designed and deployed a multi-objective ranking system on Spotify Home (music, podcasts, audiobooks), balancing relevance, diversity, and creator exposure.
- Applied heterogeneous treatment effect modeling to redesign experimentation strategy for improved detection of segment-level and long-term effects.

- Mentored interns and junior scientists, leading to publications at WWW and RecSys.

Spotify, NYC - Research Scientist (Intern)

June 2020 - August 2020

- Built and deployed a multi-objective ranking model for Home page, replacing the legacy system and improving diversity.

Pandora Media, CA - Data Scientist (Intern)

June 2017 - August 2017 & June 2018 - August 2018 (interned twice)

- Designed a model to predict user tolerance for song repetition, reducing skip rate and increasing session duration.
- Authored a poster and workshop paper at ACM RecSys 2017 presenting novel approaches to user behavior modeling in music recommendation systems.

Insight Centre for Data Analytics, Ireland - Research Assistant

April 2014 - December 2014

- Investigated consensus algorithms for group decision-making and group recommender systems, evaluating trade-offs between fairness, satisfaction, and personalization.

TOSAN CO, Iran- Software Engineer

December 2011 - December 2013

- Developed production banking and USSD systems.

TEACHING & PROFESSIONAL SERVICES

- Instructor (CU Boulder); Invited lectures (NHH, Bergen, Salamanca); PC member (RecSys, SIGIR, WSDM, WWW).

SELECTED PUBLICATIONS

- [1] **Abdollahpouri, Himan**, and Robin Burke. "Multistakeholder recommender systems." In *Recommender systems handbook*, 2021.
- [2] Jannach, Dietmar, and **Himan Abdollahpouri**. "A survey on multi-objective recommender systems." *Frontiers in Big Data* 6 (2023).
- [3] Saito, Yuta, **Himan Abdollahpouri** et al. "Long-term Off-Policy Evaluation and Learning." In *Proceedings of the ACM on Web Conference 2024*.
- [4] **Abdollahpouri, Himan** et al. "Calibrated Recommendations as a Minimum-Cost Flow Problem." In *Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining (WSDM)*, 2023.
- [5] **Abdollahpouri, Himan** et al. "User-centered evaluation of popularity bias in recommender systems." In *Proceedings of the 29th ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*, 2021.